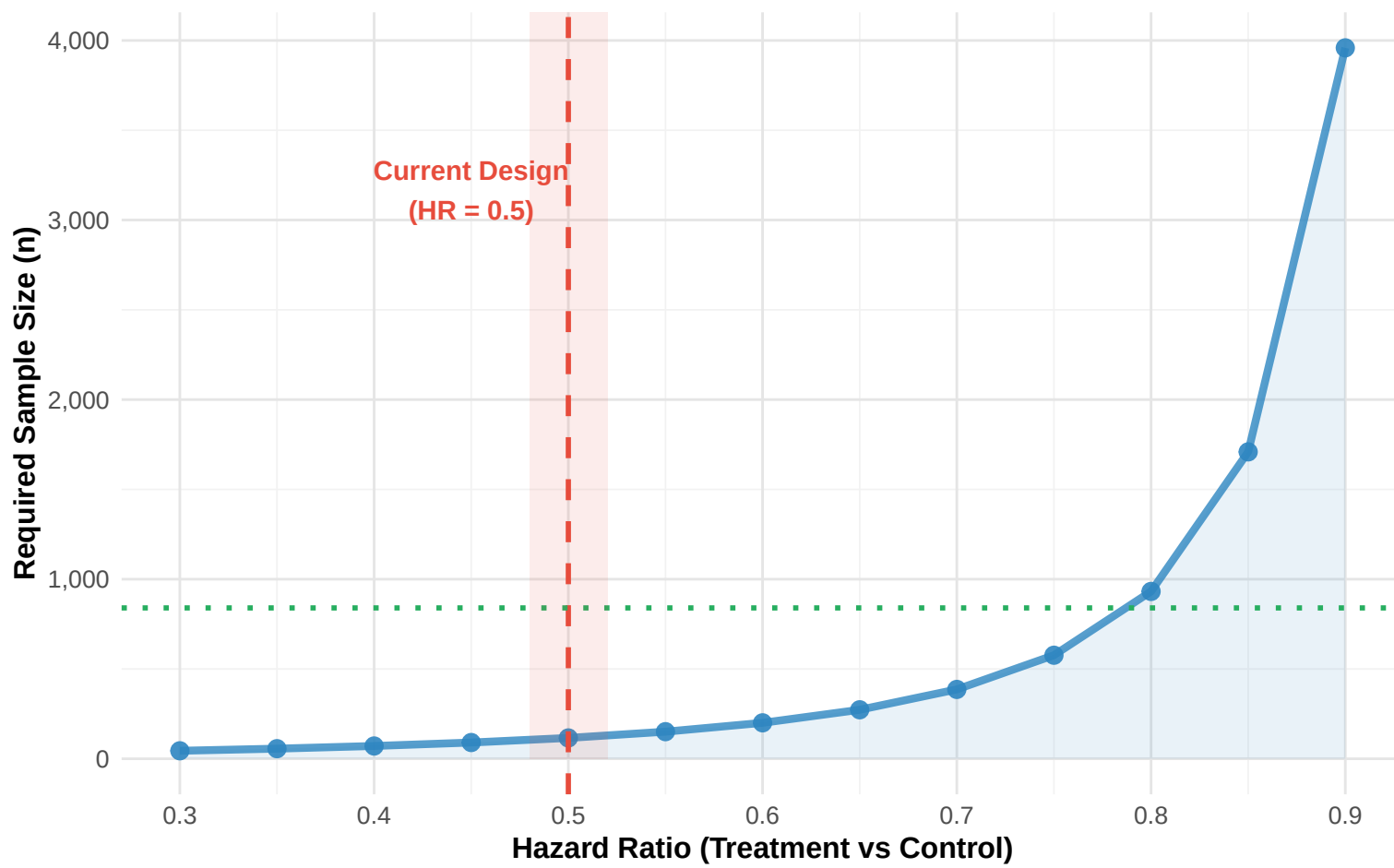


# ALONE-AF Trial: Comprehensive Power Analysis

## Sample Size Requirements vs Hazard Ratio

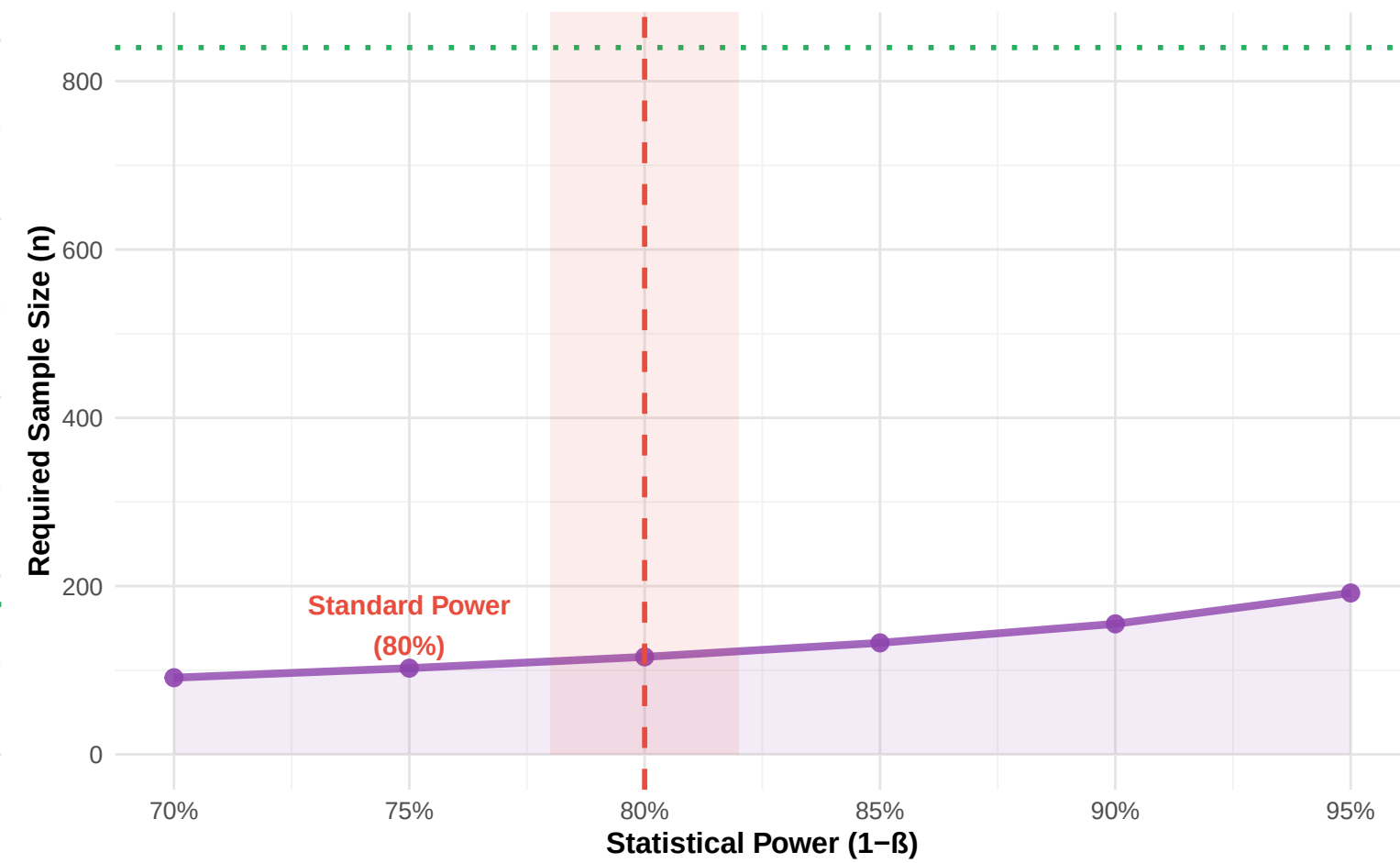
Control Event Rate = 20%, Power = 80%,  $\alpha$  = 0.05, Dropout = 6%



Dashed line: Current design (HR = 0.5)  
Dotted line: ALONE-AF actual design (n = 840)

## Sample Size Requirements vs Statistical Power

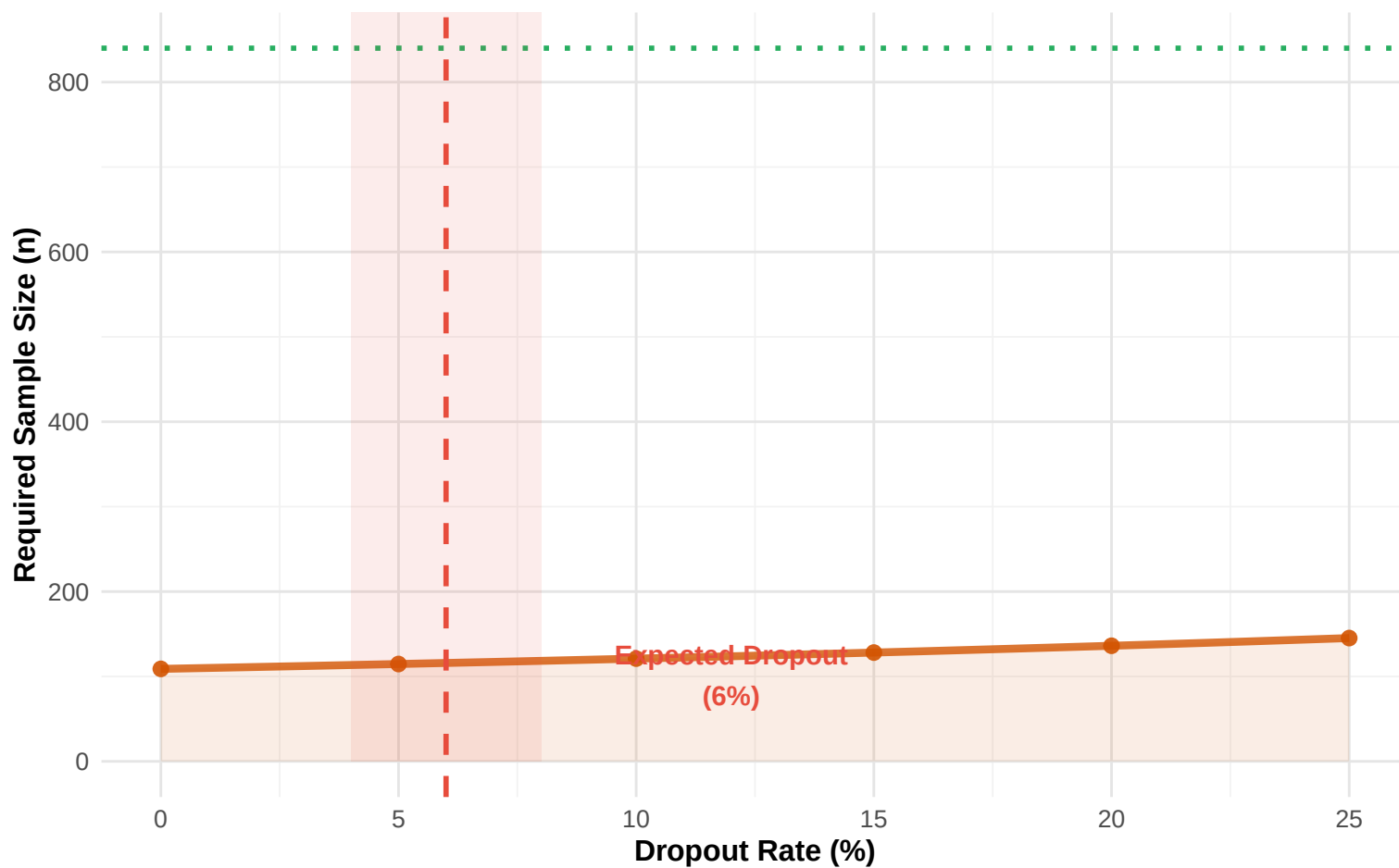
HR = 0.5, Control Rate = 20%,  $\alpha$  = 0.05, Dropout = 6%



Dashed line: Standard power (80%)  
Dotted line: ALONE-AF actual design (n = 840)

## Sample Size Requirements vs Dropout Rate

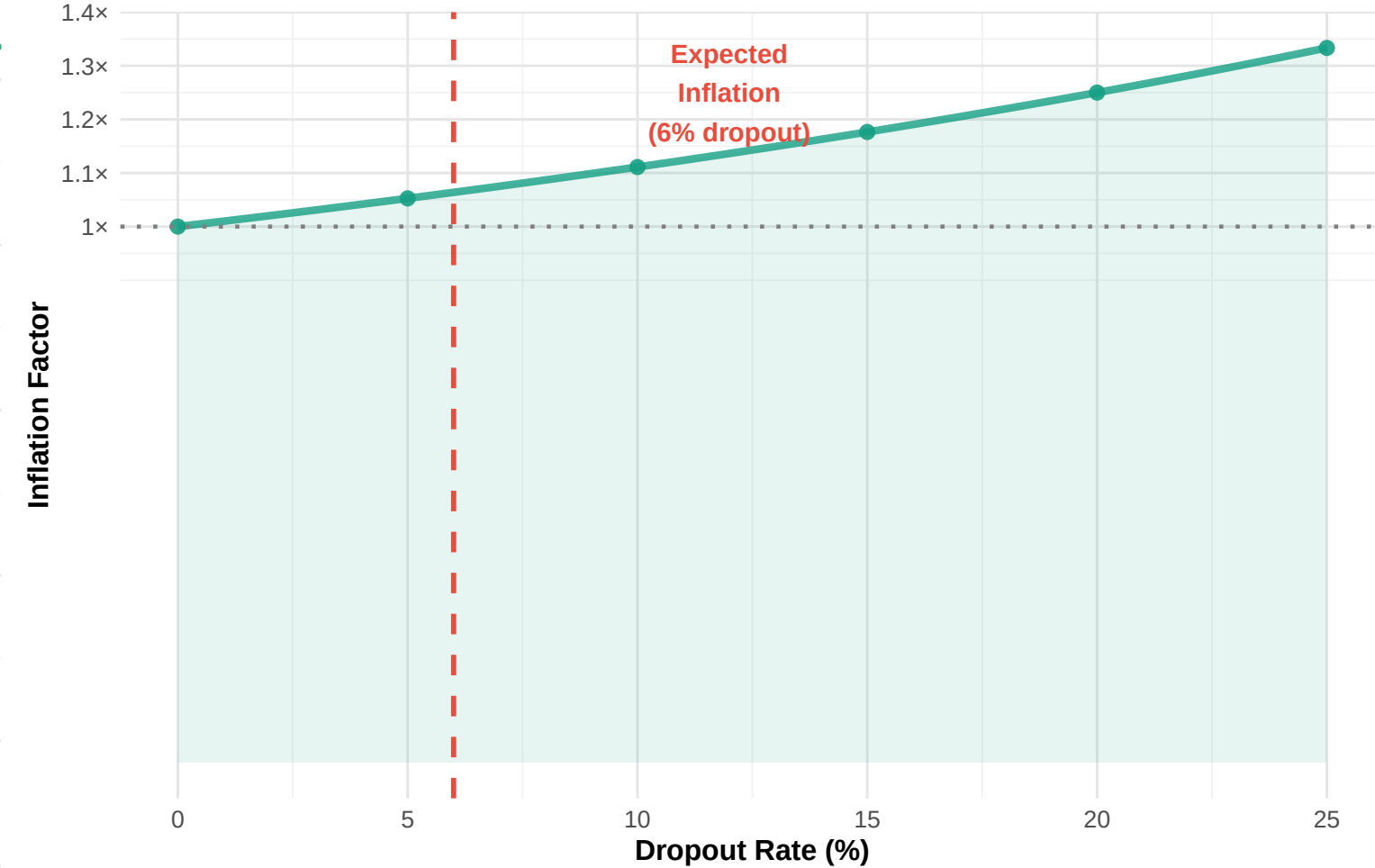
HR = 0.5, Control Rate = 20%, Power = 80%



Dashed line: Expected dropout (6%)  
Dotted line: ALONE-AF actual design (n = 840)

## Sample Size Inflation Factor vs Dropout Rate

Multiplicative effect of dropout on sample size requirements



Factor = Required Sample Size / Base Sample Size (no dropout)

## ALONE–AF Trial Design Summary

### Key Findings:

- Theoretical requirement: 116 patients (80% power, 6% dropout)
- ALONE–AF design: 840 patients (96.5% power achieved)
- Design provides 7.3× safety margin for statistical power
- Can detect minimum HR of 0.598 (40% risk reduction)
- Robust against various assumptions and parameter changes

### Design Advantages:

- Conservative and realistic event rate assumptions (20% vs 10%)
- Adequate protection against Type I and Type II errors
- Sufficient power to detect clinically meaningful differences
- Robust design accommodating potential parameter variations
- High confidence in study conclusions and clinical applicability